Year 1 AS Pure Maths

Chapter 1

Algebraic expressions

Practice Booklet

Contents

1.1 Index laws ..... book p. 1
1.2 Expanding brackets ..... book p. 2
1.3 Factorising ..... book p. 3
1.4 Negative and fractional indices ..... book p. 3
1.5 Surds ..... book p. 4
1.6 Rationalising denominators ..... book p. 5
Problem solving: Set A ..... book p. 6
Problem solving: Set B ..... book p. 6
Selected exam question bank questions .....
Answers .....

1.1 Index laws

Practice Book, printed page 1.

1. Simplify these expressions:

a. $b^3 \times b^4 \times b^2$

b. $\dfrac{a^5}{a^3}$

c. $5x^7 \times 3x^2$

d. $(2x^2)^3 \div 4x^5$

Hint

Use the laws of indices to simplify powers of the same base:

$$a^m \times a^n = a^{m+n} \qquad (a^m)^n = a^{mn}$$

$$a^m \div a^n = a^{m-n} \qquad (ab)^n = a^n b^n$$

2. Expand these expressions and simplify if possible:

a. $x^3(4x^2 - 7) + 2x^5$

b. $-5x^2(3 - 8x)$

c. $x(3x + 4) - 7(5x - 2)$

Hint

A minus sign before a bracket changes the signs of the terms inside the brackets.

3. Simplify these expressions:

a. $\dfrac{x^9 - x^5 - x^7}{x^3}$

b. $\dfrac{12x^3 + 8x^7}{4x}$

c. $\dfrac{6x^4 + 12x^3 - (4x^5)^2}{2x^3}$

Hint

Divide each term in the numerator by the denominator.
Remember that $a^1 = a$ and $a^0 = 1$

4. Simplify these expressions:

a. $3x^4y^2 \times 7xy^3$

b. $\dfrac{25x^{12}y^2}{5x^3y}$

Hint

Simplify the numbers first and then use the laws of indices to simplify each letter.

5. E Simplify these expressions:

a. $\dfrac{27r^5s^7}{3r^3}$ (2 marks)

b. $\dfrac{2(4a^4b^2)^3}{8ab^2}$ (2 marks)

6. E/P

a. Given that $32 = 2^a$, find the value of a. (1 mark)

b. Given that $81 = 3^b$, find the value of b. (1 mark)

7. E/P Given that $\dfrac{33x^6 - 3(xy)^4 + (3x^2)^3}{6x^2}$ can be written in the form $10x^p - \dfrac{1}{2}x^qy^r$, find the values of p, q and r. (3 marks)

8. E Write down the value of:

a. $(125x^3)^{1/3}$ (2 marks)

b. $\dfrac{20x^{5/4}}{5x^{1/4}}$ (2 marks)

c. $(xy)^0$ (1 mark)

1.2 Expanding brackets

Practice Book, printed page 2.

1. Expand these expressions and simplify if possible:

a. $(x + 3)(x + 4)$

b. $(x + 7)(x - 7)$

c. $(x - 6)(x - 3)$

Hint

Multiply each term in one expression by each term in the other expression and then simplify. A minus sign in front of a term will change the sign of the term it is multiplied with.

2. Expand these expressions and simplify if possible:

a. $(x + y)^2$

b. $(x - 4)^2$

c. $(2x - 5y)^2$

Hint

$(a + b)^2 = (a + b)(a + b)$

3. Expand these expressions and simplify if possible:

a. $(5x + 3y)(4x - 7y)$

b. $(x - 9y)(x^2 - 1)$

c. $(x + y)(x - 3y + 2)$

Hint

Multiply each term in one expression by each term in the other expression and then simplify. A minus sign in front of a term will change the sign of the term it is multiplied with.

4. Expand these expressions and simplify if possible:

a. $x(2x - 4y)(5x - 3y - 9)$

b. $(3x - 2y)^3$

c. $(2x - 1)(x + 3)(4x - 2)$

Hint

You can expand three brackets by expanding and simplifying one pair of brackets first, then multiplying your expanded expression by every term in the third bracket.

5. E Expand and simplify $(a + b)^3$. (2 marks)

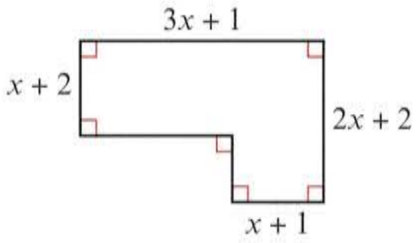
6. E/P Given that $(2x-7)(ax+b) = 6x^2 - 13x - 28$, find the values of the constants a and b. (2 marks)

7. E The length of each edge of a cube is $(2x + 1)$ cm. Find an expression in terms of x for:

a. the area of one face of the cube (2 marks)

b. the volume of the cube. (2 marks)

8. E/P The dimensions of a patio are shown, with lengths given in metres. The six edges are straight lines.



a. Find an expression for the area of the patio in terms of x. (3 marks)

b. Given that the area is 62 m^2, find the value of x. (3 marks)

9. E/P Given that $(5a - b)(3a + 2b)(2a - b) = pa^3 - qa^2b - rab^2 + sb^3$, find the values of the constants p, q, r and s. (2 marks)

1.3 Factorising

Practice Book, printed page 3.

1. Factorise these expressions completely:

a. $6x + 18$

b. $27x^2 - 9x$

c. $x^3 - 4x^2$

Hint

Find the highest common factor of the terms and write this factor outside the brackets. You can check your answers by expanding the brackets.

2. Factorise these expressions completely:

a. $7ab^2 + 21a^2b^2$

b. $8ab - 64b^3$

c. $5ab^4 + 20a^3b^2c + 15b^5c^2$

Hint

Find the highest common numerical factor, the highest common power of a and the highest common power of b. Then write these outside the brackets.

3. Factorise:

a. $x^2 + 5x + 6$

b. $x^2 - 3x - 10$

c. $x^2 - 49$

d. $y^2 - 16x^2$

Hint

These are all quadratic expressions. They can be factorised into two brackets. For parts **c** and **d** you can use the rule for the difference of two squares:

$$x^2 - y^2 = (x + y)(x - y)$$

4. Factorise completely:

a. $3x^4 + 6x^3 - 18x^2$

b. $4x^2 + 28x + 48$

c. $2x^2 + x - 3$

Hint

Start by writing part **b** as $4(x^2 + 7x + 12)$, then factorise the expression inside the brackets.

5. E Factorise completely $x - 25x^3$. (2 marks)

6. E Factorise completely $36x - 16x^3$. (2 marks)

7. E Factorise completely $8x^3 + 20x^2 + 8x$. (2 marks)

8. E/P Simplify $\dfrac{2x^3 + 6x^2 + 4x}{x^2 + 4x + 3}$. (3 marks)

1.4 Negative and fractional indices

Practice Book, printed page 3.

1. Simplify:

a. $x^{-4} \times x^{-1}$

b. $\dfrac{x^5}{x^6}$

c. $3x^{-10} \div 6x^3$

Hint

Use the rules $a^m \times a^n = a^{m+n}$ and $a^m \div a^n = a^{m-n}$

2. Simplify:

a. $2x^{1/4} \times 5x^{2/3}$

b. $9x^{5/2} \div 3x^{2/3}$

c. $(8x^3)^{2/3}$

d. $7x^{-0.25} \times 2x^{0.5}$

Hint

You can use the laws of indices with fractional powers.

3. Simplify:

a. $\sqrt[5]{x} \times \sqrt{x}$

b. $\sqrt[4]{x} \div (\sqrt{x})^3$

c. $(\sqrt[3]{x})^2 \times (\sqrt{x})^7$

Hint

Rewrite the questions with fractional powers using $a^{\frac{1}{m}} = \sqrt[m]{a}$ and $a^{\frac{n}{m}} = \sqrt[m]{a^n}$, and then simplify.

4. Evaluate:

a. $16^{3/2}$

b. $(-4)^{-3}$

c. $\left(\frac{125}{27}\right)^{-2/3}$

Hint

For parts **b** and **c**, you can use $a^{-m} = \frac{1}{a^m}$
You can use your calculator to evaluate negative and fractional powers.

5. Simplify:

a. $\left(\frac{16}{25}x^3\right)^{5/2}$

b. $\frac{x^5 + 8x^3}{x^{10}}$

c. $\frac{30x^2 - 18x^5}{6x^7}$

Hint

For part **a**, apply the power to both the fraction and x^3.

6. E

a. Evaluate $64^{3/2}$. (2 marks)

b. Simplify fully $x^3(2x^{1/3})^6$. (2 marks)

7. E

a. Find the value of $25^{-1/2}$. (2 marks)

b. Simplify $x\left(\frac{3}{2}x^{-1/4}\right)^8$. (2 marks)

8. E/P

a. Evaluate $81^{3/4}$, giving your answer as an integer. (2 marks)

b. Simplify fully $\frac{(3x^{1/2})^3}{9x^2}$. (3 marks)

9. E/P Given that $y = \frac{1}{9}x^2$, express each of the following in the form kx^n, where k and n are constants.

a. $y^{1/2}$ (2 marks)

b. $2y^{-1}$ (2 marks)

10. E/P Express 25^{2x-5} in the form 5^y, where $y = ax + b$ for some constants a and b to be determined. (2 marks)

1.5 Surds

Practice Book, printed page 4.

1. Simplify:

a. $\sqrt{18}$

b. $\sqrt{63}$

c. $\sqrt{250}$

Hint

You can manipulate surds using the rules $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ and $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

To simplify, look for factors of each number that are square numbers.

2. Simplify:

a. $\frac{\sqrt{27}}{3}$

b. $\frac{\sqrt{98}}{7}$

c. $\frac{\sqrt{24}}{2}$

Hint

Simplify the numerator first, then check whether the denominator is a factor of the numerator. If so, you can divide through by it.

3. Simplify:

a. $4\sqrt{32} - 3\sqrt{8}$

b. $\sqrt{75} + 2\sqrt{12} - \sqrt{27}$

c. $\sqrt{200} + \sqrt{18} - 2\sqrt{72}$

Hint

Simplify each surd, then collect like terms.

4. Expand and simplify if possible:

a. $\sqrt{3}(\sqrt{27} - 1)$

b. $(1 + \sqrt{2})(3 - 2\sqrt{2})$

c. $(4 - \sqrt{3})(6 - \sqrt{7})$

Hint

Multiply each term in one expression by each term in the other expression.

5. E

a. Simplify $\sqrt{20} + \sqrt{45}$, giving your answer in the form $a\sqrt{b}$ where a and b are integers. (2 marks)

b. Express $\sqrt{112}$ in the form $a\sqrt{7}$, where a is an integer. (1 mark)

6. E/P Solve the equation $x - \sqrt{60} = 2\sqrt{3} - x$, giving your answer in the form $\sqrt{a} + \sqrt{b}$, where a and b are integers. (2 marks)

7. E Expand and simplify:

a. $(2 - \sqrt{7})(\sqrt{7} - 1)$ (3 marks)

b. $(2\sqrt{7} + 3)^2$ (2 marks)

8. E Simplify:

a. $(3\sqrt{11})^2$ (1 mark)

b. $(7 + \sqrt{3})(2 - \sqrt{3})$ (3 marks)

9. E/P Given that $243\sqrt{3} = 3^a$, find a. (2 marks)

1.6 Rationalising denominators

Practice Book, printed page 5.

1. Simplify:

a. $\dfrac{1}{\sqrt{3}}$

b. $\dfrac{35}{\sqrt{5}}$

c. $\dfrac{9}{3\sqrt{3}}$

Hint

For fractions in the form $\dfrac{1}{\sqrt{a}}$ multiply numerator and denominator by $\sqrt{a}$

2. Simplify:

a. $\dfrac{\sqrt{7}}{\sqrt{21}}$

b. $\dfrac{\sqrt{12}}{\sqrt{72}}$

c. $\dfrac{\sqrt{75}}{\sqrt{125}}$

Hint

Multiply numerator and denominator by the surd denominator and then simplify.

3. Rationalise the denominator and simplify:

a. $\dfrac{1}{1+\sqrt{2}}$

b. $\dfrac{2}{\sqrt{7}-1}$

c. $\dfrac{3}{\sqrt{6}-\sqrt{5}}$

Hint

For fractions in the form $\dfrac{1}{a+\sqrt{b}}$ multiply the numerator and denominator by $a-\sqrt{b}$

For fractions in the form $\dfrac{1}{a-\sqrt{b}}$ multiply the numerator and denominator by $a+\sqrt{b}$

4. Rationalise the denominator and simplify:

a. $\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

b. $\dfrac{3-\sqrt{5}}{\sqrt{5}+5}$

c. $\dfrac{3}{(2\sqrt{2}+3)^2}$

Hint

For part **c**, start by expanding and simplifying the denominator.

5. E Express $\dfrac{2}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are integers to be found. (2 marks)

6. E Write each expression in the form $a + b\sqrt{5}$, where a and b are integers to be found:

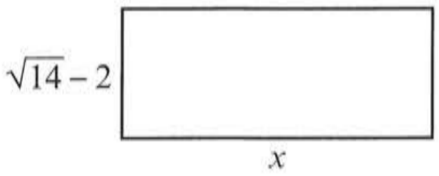
a. $\dfrac{1+\sqrt{5}}{\sqrt{5}-2}$ (3 marks)

b. $\dfrac{7+\sqrt{5}}{3+\sqrt{5}}$ (3 marks)

7. E/P $\dfrac{3x^5 - x^{5/2}}{\sqrt{x}}$ can be written in the form $3x^p - x^q$. Write down the values of p and q. (2 marks)

8. E/P Show that $\dfrac{(7-2\sqrt{x})^2}{\sqrt{x}}$ can be written in the form $Ax^{-1/2} + Bx^{1/2} - C$ where A, B and C are constants to be found. (3 marks)

9. E/P The diagram shows a rectangle with sides of length $(\sqrt{14} - 2)$ cm and x cm. The area of the rectangle is 5 cm^2. Find x, giving your answer as simply as possible in surd form.



(3 marks)

Problem solving: Set A

Practice Book, printed page 6.

Bronze

a. Simplify $\sqrt{147} - \sqrt{75}$ giving your answer in the form $a\sqrt{3}$, where a is an integer. (2 marks)

b. Hence, or otherwise, simplify $\dfrac{24\sqrt{2}}{\sqrt{147} - \sqrt{75}}$, giving your answer in the form $b\sqrt{6}$, where b is an integer. (2 marks)

Silver

a. Expand and simplify $(11+\sqrt{5})(\sqrt{5}+1)$ giving your answer in the form $a+b\sqrt{5}$, where a and b are integers. (2 marks)

b. Hence, or otherwise, simplify $\dfrac{11+\sqrt{5}}{\sqrt{5}-1}$, giving your answer in the form $c+d\sqrt{5}$, where c and d are integers. (2 marks)

Gold

Simplify $\dfrac{6\sqrt{3}-4}{5-\sqrt{3}}$, giving your answer in the form $p\sqrt{3} - q$, where p and q are positive rational numbers. (4 marks)

Problem solving: Set B

Practice Book, printed page 6.

Bronze

a. Express 4^{x+2} in the form 2^y, stating y in terms of x. (2 marks)

b. Hence, or otherwise, solve the equation $4^{x+2} = 32$. (2 marks)

Silver

a. Rewrite the equation $(4^{x-1})^2 - 6(4^{x-1}) + 8 = 0$ in the form $y^2 + by + c = 0$, where $y = 4^{x-1}$ and b and c are constants to be found. (1 mark)

b. Factorise your equation from part **a** and hence, or otherwise, solve for x. You must show each step of your working. (3 marks)

Gold

Solve the following equation, showing each step of your working:

$$(9^{x-1})^2 - 30(9^{x-1}) + 81 = 0$$

(5 marks)

Selected exam question bank questions

Question 1

a. Find the value of $27^{5/3}$. (2 marks)

b. Simplify fully $\dfrac{(4x^{1/2})^3}{8x^2}$. (3 marks)

Question 2

Simplify $\dfrac{4\sqrt{3}-2}{7-\sqrt{3}}$, giving your answer in the form $p\sqrt{3}-q$, where p and q are positive rational numbers to be found. (4 marks)

Question 4

a. Simplify $\sqrt{98}-\sqrt{50}$ giving your answer in the form $a\sqrt{2}$, where a is an integer to be found. (2 marks)

b. Hence, or otherwise, simplify $\dfrac{16\sqrt{3}}{\sqrt{98}-\sqrt{50}}$, giving your answer in the form $b\sqrt{c}$, where b and c are integers and $b \neq 1$. (3 marks)

Question 24

a. Factorise completely $x^3 + 12x^2 + 36x$. (2 marks)

b. Sketch the curve with equation $y = x^3 + 12x^2 + 36x$, showing the coordinates of the points where the curve cuts or touches the x-axis. (2 marks)

Question 36

Solve the following equation algebraically, showing each step of your working:

$$(4^{x-1})^2 - 10(4^{x-1}) + 16 = 0.$$

(5 marks)

Question 46

The path of a toy rocket launched into the air can be modelled by the equation $y = x - 0.05x^2$, where x is the horizontal distance the rocket travels (in metres) and y is the vertical height (in metres) measured from where the rocket was fired.

a. Sketch the graph of $y = x - 0.05x^2$ to show the path of the rocket. (2 marks)

b. State the horizontal distance travelled by the rocket when it lands on the ground. (3 marks)

In reality the rocket achieves a maximum height of 4.5 m.

c. Evaluate the model in light of this information. (2 marks)

Question 64

a. Prove that, if $1 + x^2 < (1 + x)^2$, then $x > 0$. (3 marks)

b. Show, by means of a counter-example, that the inequality $1 + x^2 < (1 + x)^2$ is not true for all values of x. (2 marks)

Answers

1.1 Index laws

1. a. b^9; b. a^2; c. $15x^9$; d. $2x$.

2. a. $6x^5 - 7x^3$; b. $-15x^2 + 40x^3$; c. $3x^2 - 31x + 14$.

3. a. $x^6 - x^2 - x^4$; b. $3x^2 + 2x^6$; c. $3x + 6 - 8x^7$.

4. a. $21x^5y^5$; b. $5x^9y$.

5. a. $9r^2s^7$; b. $16a^{11}b^4$.

6. a. 5; b. 4.

7. $p = 4$, $q = 2$, $r = 4$.

8. a. $5x$; b. $4x$; c. 1.

1.2 Expanding brackets

1. a. $x^2 + 7x + 12$; b. $x^2 - 49$; c. $x^2 - 9x + 18$.

2. a. $x^2 + 2xy + y^2$; b. $x^2 - 8x + 16$; c. $4x^2 - 20xy + 25y^2$.

3. a. $20x^2 - 23xy - 21y^2$; b. $x^3 - 9x^2y - x + 9y$; c. $x^2 + 2x + 2y - 2xy - 3y^2$.

4. a. $10x^3 - 18x^2 - 26x^2y + 12xy^2 + 36xy$; b. $27x^3 - 54x^2y + 36xy^2 - 8y^3$; c. $8x^3 + 16x^2 - 22x + 6$.

5. $a^3 + b^3 + 3a^2b + 3ab^2$.

6. $a = 3$, $b = 4$.

7. a. $(4x^2 + 4x + 1)$ cm^2; b. $(8x^3 + 12x^2 + 6x + 1)$ cm^3.

8. a. $(4x^2 + 8x + 2)$ m^2; b. $x = 3$.

9. $p = 30$, $q = 1$, $r = 11$, $s = 2$.

1.3 Factorising

1. a. $6(x + 3)$; b. $9x(3x - 1)$; c. $x^2(x - 4)$.

2. a. $7ab^2(1 + 3a)$; b. $8b(a - 8b^2)$; c. $5b^2(ab^2 + 4a^3c + 3b^3c^2)$.

3. a. $(x + 3)(x + 2)$; b. $(x - 5)(x + 2)$; c. $(x + 7)(x - 7)$; d. $(y + 4x)(y - 4x)$.

4. a. $3x^2(x^2 + 2x - 6)$; b. $4(x + 3)(x + 4)$; c. $(2x + 3)(x - 1)$.

5. $x(1 - 5x)(1 + 5x)$.

6. $4x(3 + 2x)(3 - 2x)$.

7. $4x(2x + 1)(x + 2)$.

8. $\dfrac{2x(x + 2)}{x + 3}$.

1.4 Negative and fractional indices

1. a. x^{-5}; b. x^{-1}; c. $\dfrac{1}{2}x^{-13}$.

2. a. $10x^{11/12}$; b. $3x^{11/6}$; c. $4x^2$; d. $14x^{0.25}$.

3. a. $x^{7/10}$; b. $x^{-5/4}$; c. $x^{25/6}$.

4. a. 64; b. $-\dfrac{1}{64}$; c. $\dfrac{9}{25}$.

5. a. $\dfrac{1024}{3125}x^{15/2}$; b. $x^{-5} + 8x^{-7}$; c. $5x^{-5} - 3x^{-2}$.

6. a. 512; b. $64x^5$.

7. a. $\dfrac{1}{5}$; b. $\dfrac{6561}{256}x^{-1}$.

8. a. 27; b. $3x^{-1/2}$.

9. a. $\frac{1}{3}x$; b. $18x^{-2}$.

10. 5^{4x-10}.

1.5 Surds

1. a. $3\sqrt{2}$; b. $3\sqrt{7}$; c. $5\sqrt{10}$.

2. a. $\sqrt{3}$; b. $\sqrt{2}$; c. $\sqrt{6}$.

3. a. $10\sqrt{2}$; b. $6\sqrt{3}$; c. $\sqrt{2}$.

4. a. $9-\sqrt{3}$; b. $-1+\sqrt{2}$; c. $24-4\sqrt{7}-6\sqrt{3}+\sqrt{21}$.

5. a. $5\sqrt{5}$; b. $4\sqrt{7}$.

6. $\sqrt{15}+\sqrt{3}$.

7. a. $-9+3\sqrt{7}$; b. $37+12\sqrt{7}$.

8. a. 99; b. $11-5\sqrt{3}$.

9. $a=\frac{11}{2}$.

1.6 Rationalising denominators

1. a. $\frac{\sqrt{3}}{3}$; b. $7\sqrt{5}$; c. $\sqrt{3}$.

2. a. $\frac{\sqrt{3}}{3}$; b. $\frac{\sqrt{6}}{6}$; c. $\frac{\sqrt{15}}{5}$.

3. a. $\sqrt{2}-1$; b. $\frac{1+\sqrt{7}}{3}$; c. $3\sqrt{6}+3\sqrt{5}$.

4. a. $5+2\sqrt{6}$; b. $\frac{5-2\sqrt{5}}{5}$; c. $51-36\sqrt{2}$.

5. $4+2\sqrt{3}$.

6. a. $7+3\sqrt{5}$; b. $4-\sqrt{5}$.

7. $p=\frac{9}{2}$, $q=2$.

8. $A=49$, $B=4$, $C=28$.

9. $\frac{2+\sqrt{14}}{2}$.

Problem solving: Set A

Bronze. a. $2\sqrt{3}$; b. $4\sqrt{6}$.

Silver. a. $16+12\sqrt{5}$; b. $4+3\sqrt{5}$.

Gold. $\frac{13}{11}\sqrt{3}-\frac{1}{11}$.

Problem solving: Set B

Bronze. a. 2^{2x+4}; b. $x=\frac{1}{2}$.

Silver. a. $y^2-6y+8=0$; b. $x=\frac{3}{2}$ or $x=2$.

Gold. $x=\frac{3}{2}$ or $x=\frac{5}{2}$.

Selected exam question bank answers

Question 1

a. 243
b. $8x^{-1/2}$

Question 2

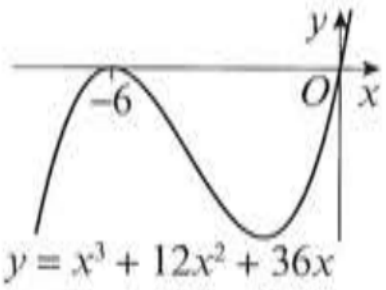
$\frac{13}{23}\sqrt{3} - \frac{1}{23}$

Question 4

a. $2\sqrt{2}$
b. $4\sqrt{6}$

Question 24

a. $x(x+6)(x+6)$
b.

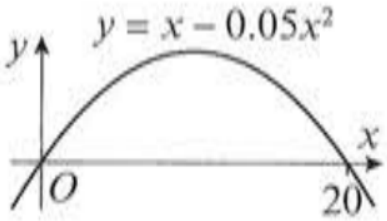


Question 36

$x = \frac{3}{2}$ or $x = \frac{5}{2}$

Question 46

a.



b. 20 m
c. Model gives maximum height of 5 m

Question 64

a. $1 + x^2 < 1 + x^2 + 2x \Rightarrow 0 < 2x \Rightarrow x > 0$
b. For $x = -1$, $1 + (-1)^2 < (1 + (-1))^2 \Rightarrow 2 < 0$ (not true)